

Data and Visualization of COVID-19

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Executive Summary: [¼ page description of the project highlights. The summary needs to answer the following questions: What is the problem you are working on? Why is it important? What are the key results and how are they useful to the reader?]

➤ Executive Summary

During the COVID-19 outbreak between early 2020 and mid-2023, Texas lawmakers had to make difficult decisions to mitigate the virus's impact while balancing public health and economic concerns. This project analyzes the effectiveness of two primary interventions—social distancing measures and vaccination campaigns—in reducing infection rates, hospitalizations, and mortality. By examining county and state-level data, we assess how different regions were affected and identify the most effective strategies for controlling the spread of the virus. The findings highlight that while social distancing initially helped slow infection rates, its effects were temporary and often reversed once restrictions were lifted. In contrast, vaccination had a more sustained impact, significantly lowering hospitalizations and deaths. The results suggest that future pandemic responses should prioritize rapid vaccine distribution, booster campaigns, and healthcare accessibility over prolonged social distancing measures. By using real-time data to study the changes of COVID-19, Texas can ensure a more effective and sustainable approach to managing future health crises.

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1 Problem Description

COVID-19, caused by the SARS-CoV-2 virus, is highly contagious and has caused an international public health crisis. The virus is transmitted primarily through respiratory droplets and its symptoms can be mild flu-like manifestations to severe respiratory distress and death. Due to its rapid rate of spread and the fact that its variants are constantly evolving, COVID-19 has placed immense pressure on healthcare systems all over the world.

One of the primary ways to limit the spread of COVID-19 has been social distancing—physical space between individuals to reduce the spread. The Centers for Disease Control and Prevention (CDC) has recommended a minimum of six feet of space in public places. The purpose of these measures is to "flatten the curve," reducing the rate of infection so as not to overwhelm hospitals. Mask mandates and vaccination drives are some other interventions that have played significantly in controlling outbreaks. To create effective policies, it is important to examine data regarding virus transmission, hospitalization, and resources. Through this examination, policymakers can evaluate risks, distribute resources effectively, and make timely interventions to reduce the effects of the pandemic. By knowing how cases progress in various areas, policymakers can create strategies that maximize public safety while reducing unnecessary economic or social disruptions.

One of the main issues of this analysis is measuring the comparative worth of different public health interventions. In specific, policymakers need to answer:

- How has social distancing affected COVID-19 infection rates compared to the impact of vaccination campaigns? Having knowledge of the impact of each intervention on infection rates will tell policymakers whether in the future policy should concentrate on distancing interventions, vaccination campaigns, or both.
- Which of the two interventions—social distancing or vaccination—had greater impact in reducing hospitalization and mortality? This is crucial to optimize resource utilization and avoid overloading healthcare systems.
- If someone has gotten infected, how good are their chances of survival due to the vaccine? By comparing fatality and hospitalization rates among vaccinated and unvaccinated groups, policymakers can assess the vaccine's potential to prevent severe illness and guide future public health messaging.

The most important stakeholders of this analysis are state-level policymakers in the United States, who have the responsibility to implement public health measures and offer resources to address the COVID-19 pandemic. These policymakers must balance protecting public health with economic and social stability. By analyzing patterns of COVID-19 cases, hospitalization rates, and the effectiveness of social distancing and vaccination, policymakers can make data-driven decisions about when to implement or relax restrictions. In

addition, pinpointing regions that are most vulnerable or most effective at containing the virus can help policymakers tailor responses to specific state and county needs. With limited healthcare resources, such as hospital beds and medical personnel, state officials must use data to allocate assistance effectively, with high-risk areas receiving the support they require. This project aims to provide key data that will benefit state-level decision-makers to improve emergency response plans as well as inform well-informed, evidence-based public health policies.

2 Data Collection and Data Quality

➤ Data Gathering

Data on mobility and specific data on deaths per number of people come from the Google Cloud Public Marketplace, which provides many publicly available datasets hosted on Google Cloud. These datasets cover a wide range of topics, including data on COVID-19, economic indicators and geographic information. The data quality is expected to be high-quality, well-structured, and often curated from authoritative sources. It also includes metadata describing the data fields, timestamps, and methods of collection.

The second source used is the Google COVID-19 Open Data Repository, which is a dataset compiled by Google, aggregating COVID-19 data from multiple global sources, including governments, health organizations, and research institutions. The data is comprehensive and globally sourced, covering case counts, trends in hospitalizations, and epidemiology. Potential data quality and consistency issues might arise due to variations in reporting standards between countries, however, this will not be a problem because the scope of the data used in the analysis is narrowed down to specifically Texas. The data for Texas is supplied by the Texas Department of State Health Services. The data will be reliable as other sources used confirm its consistency.

The final source used is the CDC (Centers for Disease Control and Prevention) COVID-19 dataset includes official U.S. COVID-19 data, such as case counts, hospitalizations, deaths, and vaccination rates. This is official data reported by public health agencies. It is well-documented and follows standard reporting procedures and is verified to be accurate by the CDC. Due to the verification process, it is one of the most authoritative sources for U.S. COVID-19 data.

3 Data Exploration

➤ Data Quality

The data from these different sources combine either at the county level using the unique FIPS code, or by using state level identifiers found in the much larger dataset, although losing out on county level scope, it still gives a great insight into the overall functionality of combatting the virus.

At this point, each dataset has different column names. Common elements such as dates and counties are cleaned to have the same structure across all data. An example is removing capitals and removing the word “county” from all counties so that all that remains is the actual name. All the dates will follow a yyyy-mm-dd format, ensuring that all merges go smoothly. Some data associated with the date is aggregated into weeks, so that the visualizations are clearer and have less clutter. This decision was made because although day-to-day information is helpful, COVID-19 takes time to show results, and data does not have much change between days, however it is clearer between weeks. There are not any outliers since they can all be attributed to real-life events, such as new variants or interventions.

One table did have some strange outliers in the data, where there were negative numbers of confirmed cases or recoveries in a day. This does not make sense and needs to be cleaned. I believe that

To effectively streamline the process, four timestamps will be taken to observe the trends without being overwhelmed. These timestamps are hand picked to large events in the timeline of COVID-19, showing us causes and effects of decisions made by the United States policymakers.

- **2020-07-15:** The first major infectious wave, also known as the “Summer Wave.” WHO officially recognized the possibility of airborne transmission just one week earlier. Many countries and U.S. states began implementing mandatory mask policies as the CDC and WHO increased their guidance on wearing masks in public spaces
- **2021-01-20:** Roughly one month after the introduction of the vaccine. This date was chosen because the first vaccines require two doses given 3–4 weeks apart for full effectiveness. We can examine how the next few weeks of data look after the vaccines are in full effect to see the effectiveness of the initial vaccine.
- **2021-10-22:** Marks a key date in the U.S. vaccine rollout, as the FDA authorized booster doses for Moderna and Johnson & Johnson vaccines. Although vaccines have been released to the public for almost a year, we can see pre-booster trends, to identify the power of the initial vaccines. This also makes it more difficult to pinpoint the effectiveness of keeping individuals healthy and safe as there is now more than one variable at play. By taking this timestamp, we can evaluate the before and after trends to see how effective it is.

- **2023-05-10:** The last day of the pandemic. The next day, the World Health Organization (WHO) declared an end to COVID-19 as a global public health emergency. This day marks the final effectiveness of interventions. Here, we can assess the long-term impact of social distancing and vaccinations in controlling the virus. Also, by the last day, most of the population had been exposed to the virus either through infection or vaccination.

These time stamps highlight a dynamic interaction between the virus' spread, public health interventions, and vaccine efforts. By focusing on these key moments, we can obtain a comprehensive understanding of how strategies like mask mandates, vaccinations, and boosters shaped the course and outcome of the pandemic.

Feature	Scale of Measurement	Description
Total_pop	Ratio	Total population of individuals living in the county
Median_income	Ratio	Median income of individuals living in the county
Median_age	Ratio	Median age of individuals living in the county
New.cases	Ratio	Count of new cases confirmed after positive test on this date, on the county level
New.deaths	Ratio	Count of new deaths from a positive COVID-19 case on this date, on the county level
New_confirmed_TX	Ratio	Count of new cases confirmed after positive test on this date, on the state level
New_deceased_TX	Ratio	Count of new deaths from a positive COVID-19 case on this date, on the state level
New_recovered_TX	Ratio	Count of new recoveries from a positive COVID-19 case on this date, on the state level
First_Dose_Pct	Ratio	People with at least one Dose by State of Residence
Full_Dose_Pct	Ratio	Percent of people who have completed a primary series (have second dose of a two-dose vaccine or one dose of a single-dose vaccine) based on the jurisdiction and county where vaccine recipient lives
Full_Dose_Pct65Plus	Ratio	Percent of people ages 65+ who have completed a primary series (have second dose of a two-dose vaccine or one dose of a single-dose vaccine) based on the jurisdiction where vaccine recipient lives
Retail_and_rec_pct	Interval	Mobility trends in percentage for places like restaurants, cafes, shopping centers, theme parks museums, libraries, and movie theaters.
Work_pct	Interval	Mobility trends for showing the percentage change of people coming into work
current_hospitalized_patients	Ratio	Count of current (active) cases hospitalized after positive test to date
current_intensive_care_patients	Ratio	Count of current (active) cases admitted into ICU after a positive COVID-19 test to date

Figure 1: Feature Descriptions.

➤ Statistical Summary

The statistical summary of the major variables of the dataset are explained in Table 2. The statistics provide insights into population trends, COVID-19 trends, vaccination rates, mobility trends, and hospitalization. Since the categorical variables such as county name or the state label is not addressed in this analysis, only numeric attributes are examined based on mean, median, mode, range, and standard deviation. One of the key observations is that there is huge variation in the population size of different regions. The mode for the overall population is 4,525,519, this is most likely because it is taking the largest population and printing that, since there is no county in the set that has the exact same population. But the enormous difference between the minimum value of 1,084 and maximum value of 4,525,519 and the large standard deviation of 399,834.4 skew the distribution to the many rural counties being grouped in with the urban ones. The range of the median_income across all the counties backs this and many statistics regarding population will be skewed due to the outliers (urban counties).

COVID-19 case rates are extremely fluctuating, ranging from 0 to 89,119 new daily cases and having a high standard deviation of 1,255.17. Some places seem to have had enormous outbreaks and others minuscule if any new cases on days. New deaths range from 0 to 270 and have a mode of 0, indicating that most days did not see deaths due to low mortality in certain regions. Again, this is because of the much smaller counties not feeling the impact as COVID-19 nearly as much as urban counties as they are much less densely populated. Since the mode is 0, it would seem no new cases and deaths were reported in some regions. However, it does not necessarily suggest the lack of COVID-19 in those areas either, it could have been because of underreporting or data delays.

Vaccination data show that most of the population was fully vaccinated by the end of the pandemic, as reflected by the mode of 95% for full and first dose percentages. The range of these percentages is large but that only shows the change from the beginning to the end. The 13.44% standard deviation for first-dose vaccinations adequately illustrates this asymmetrical distribution. For the 65+ years population, we can examine the different quartiles to view how quick people aged 65+ got the full vaccine. In the beginning it just about quadrupled, showing that the older and more at-risk population got the vaccine as quickly as possible, and continued to vaccinate until the end.

The mobility data creates an interesting picture. The retail and recreation percentage change has a very large range (-95 to 197) because there were wild fluctuations in consumer behavior, likely due to lockdowns and reopening phases. There were also many historic scares where people flocked to retail to stock up on supplies for the pandemic when they could, so this data makes sense in retrospect. Similarly, work mobility changes range from -89 to 40, which means that while some places experienced full work-from-home shifts, others experienced relatively normal work activity. Hospitalization data show immense heterogeneity in current hospitalizations (4,525 to 13,812, mode 3,247) and ICU patients (1,616 to 3,655, mode 164). The large standard deviation (3,891.7 for hospitalizations and 1,046.3 for ICU cases) is a manifestation of extreme surges in healthcare demands, likely due to COVID-19 variant surges and bed capacity constraints.

Variable	Min.	1st Qu.	Median	3rd Qu.	Max.	Mode	Std Dev	Range
Total_pop	1084	8082	20082	53171	4525519	4525519	399834.4	4524435
Median_income	24794	42354	47923	54971	93645	44601	11619.06	68851
Median_age	25.80	34.60	38.35	42.90	57.10	35.6	5.839018	31.3
New.cases	0.0	2.0	12.0	59.0	89119.0	0	1255.174	89119
New.deaths	0.0	0.0	0.0	1.0	270.0	0	10.43202	270
New_confirmed_TX	0	1692	4490	9909	164902	0	11966.99	164902
New_deceased_TX	0.00	19.00	49.50	134.25	675.00	0	104.3569	675
New_recovered_TX	0	1525	5071	9731	82755	0	804.573	82755
First_Dose_Pct	22.30	44.40	50.30	59.90	99.90	95	13.43828	77.6
Full_Dose_Pct	18.30	38.60	43.90	51.80	95.00	95	11.6431	76.7
Full_Dose_Pct65 Plus	15.70	67.20	73.80	81.60	99.90	95	10.79269	84.2
Retail_and_rec_pct	-95.00	-19.00	-9.00	-1.00	197.00	-10	16.44832	282
Work_pct	-89.0	-31.0	-23.0	-15.0	40.0	-18	13.76627	129
current_hospitalized_patients	4525	5500	6474	10143	13812	3247	3891.696	13761
current_intensive_care_patients	1616	1702	1789	2722	3655	164	1046.327	3764

Figure 2: Statistical Summary.

4 Modeling and Evaluation

When comparing the spikes in COVID-19 cases and deaths over time in Texas, a clear trend emerges while both metrics follow similar patterns, the magnitude and sharpness of their peaks differ significantly. The case spikes are much sharper and more extreme, whereas the death spikes are more gradual and smoother. This suggests that while surges in infections occur rapidly, the resulting fatalities accumulate over a longer period rather than peaking all at once. An example is the early 2022 wave, where new cases surged dramatically, exceeding 400,000 weekly cases at its peak. However, the corresponding death spike, while noticeable, was much smaller in proportion, reaching just above 2,000 weekly deaths. This indicates that the fatality rate during later waves was lower compared to earlier waves, we know this surge in cases was due to the Omicron variant, which we can see in the data to be very contagious, but not as deadly as the other forms of the virus. In contrast, earlier waves in 2020 and early 2021 show a closer relationship between case and death spikes, suggesting that the virus may have been more lethal at the time, or that fewer medical interventions were available. Over time, the divergence between case and death trends highlights how medical advancements and public health measures helped mitigate the most severe outcomes of COVID-19, even during large infection surges, especially at the very end, where there is a sizable increase in cases, but not in deaths.

A line plot is ideal for visualizing time series data because it effectively captures trends, fluctuations, and long-term patterns over a continuous period. The smooth, continuous nature of a line plot makes it easy to identify peaks, declines, and recurring trends in rates. Additionally, by using a single continuous line, the visualization avoids clutter that might arise from bar charts or scatter plots, making it easier to track the rise and fall of cases over time.

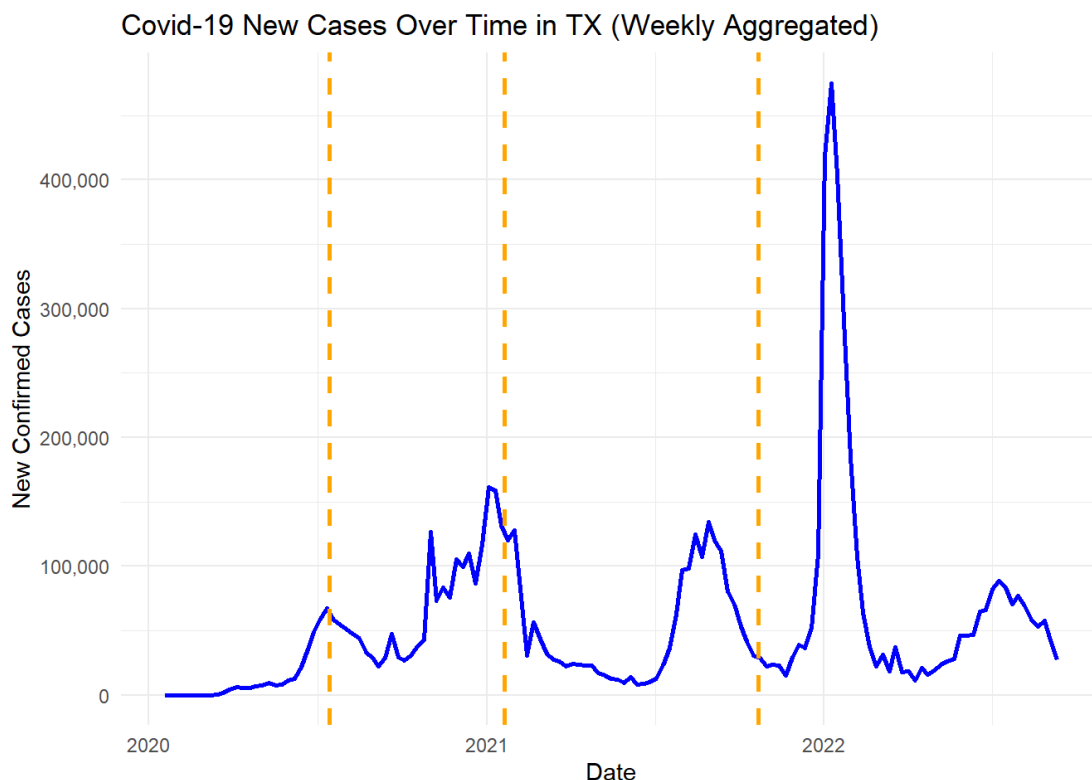


Figure 3: The new weekly COVID-19 cases recorded in Texas.

The graph illustrates the trend of weekly aggregated new confirmed COVID-19 deaths in Texas from 2020 to 2022, highlighting multiple peaks and declines over time. The red line represents the number of deaths, while the green dashed lines mark significant points, with the middle green line occurring roughly one month after the initial vaccine rollout in December 2020. The timing of the dips in deaths suggests a strong relationship between vaccination efforts and reductions in mortality.

The first major peak, occurring before early 2021, represents the winter surge before vaccines became widely available. Shortly after the vaccine rollout gained momentum, there was a significant decline in deaths, suggesting that once the vaccines were in full effect, mortality rates started to drop. However, in mid-to-late 2021, another peak emerged, likely due to the Delta variant, which caused a resurgence in cases. Following this spike, there was another sharp decline in deaths, coinciding with increased vaccination rates and broader immunity.

In late 2021 and early 2022, the Omicron variant drove another surge in deaths, though this wave was generally less severe than previous ones. Once again, a steep decline followed, reinforcing the idea that widespread

vaccination, booster doses, and improved treatments helped mitigate severe outcomes. Despite periodic resurgences, the overall trend in 2022 shows a sustained decline in deaths, indicating that COVID-19 mortality became less frequent over time, likely due to the cumulative effects of vaccination and natural immunity. This pattern suggests that while surges continue to occur, public health interventions played a critical role in reducing the severity and duration of each wave.

Using the same logic as Figure 3, Figure 4 follows the same by using the line plot. We can identify major events in the COVID-19 timeline and see how programs that were being rolled out affect the epidemic.

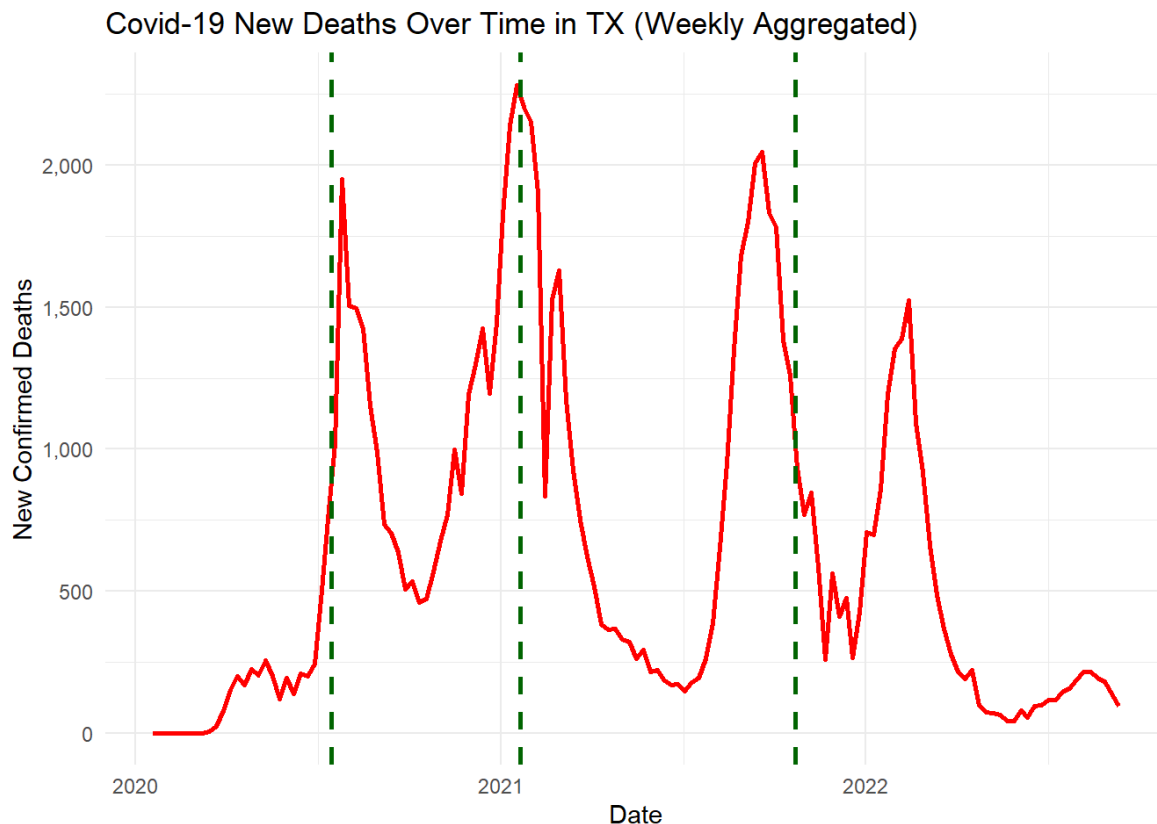


Figure 4: The new weekly deaths because of COVID-19 recorded in Texas.

A major point to note from this data is that once the vaccines went into full effect, (about a month after their distribution to the public) the number of new deaths plummeted drastically until the end of the summer. These giant spikes in deaths can be attributed to the multiple variants of the virus, each requiring its own vaccine to effectively combat. One issue with the vaccines is that they are slow to develop. If they take longer to develop that a new variant comes into rotation, then the virus can spread rapidly before an effective vaccine is available, leading to new surges in cases and deaths. This lag in vaccine development creates a window of vulnerability where the virus

can mutate further, potentially reducing the effectiveness of existing vaccines and prolonging the pandemic's impact.

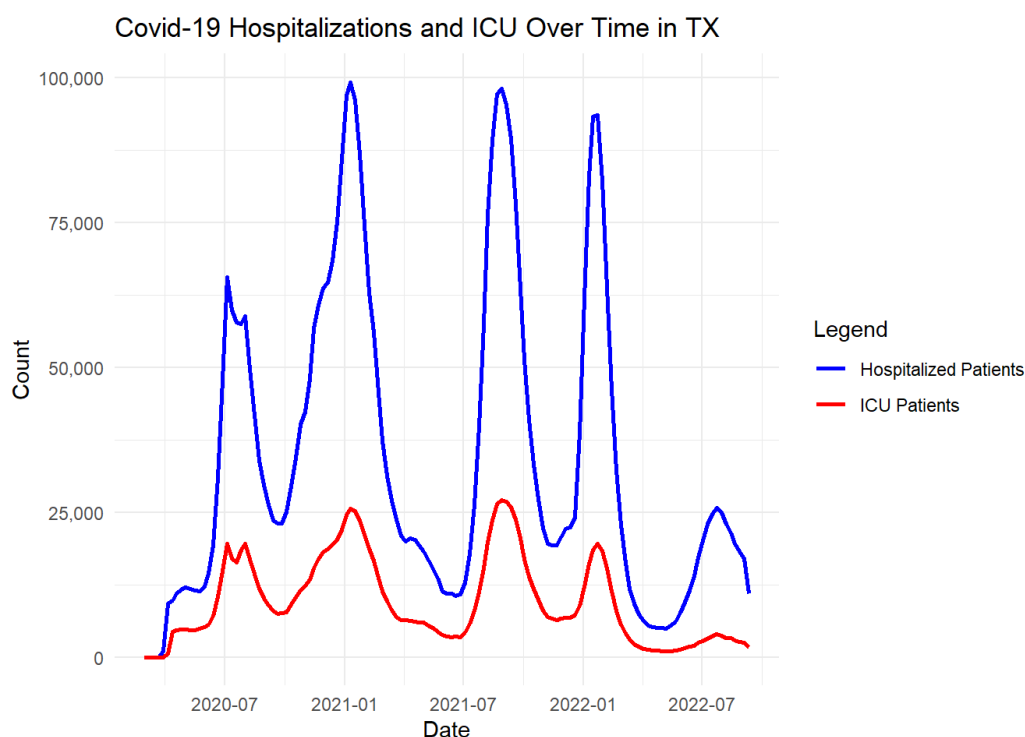


Figure 5: Visual of the number of patients in the hospital compared to the number of patients requiring intensive care.

Consistent with the other graphs, the hospitalizations follow the four spikes, but the graphs are slightly different, more resembling the deaths data over the cases data in the sense that more patients are admitted into the hospital based on how deadly the virus is. An interesting point to note is that when the Omicron variant hit, the spread was very high, but the mortality rate was low. Another trend to note is near the end of the data, where even though the number of patients admitted to the hospital began to rise again, the patients needing intensive care did not. This could be because the variants moving on were not as deadly or that the vaccines were working to the extent that people were not in dire need of care. It seems like vaccines and treatments continued to improve, the healthcare system was able to manage the increasing cases without being overwhelmed by the severity of the disease.

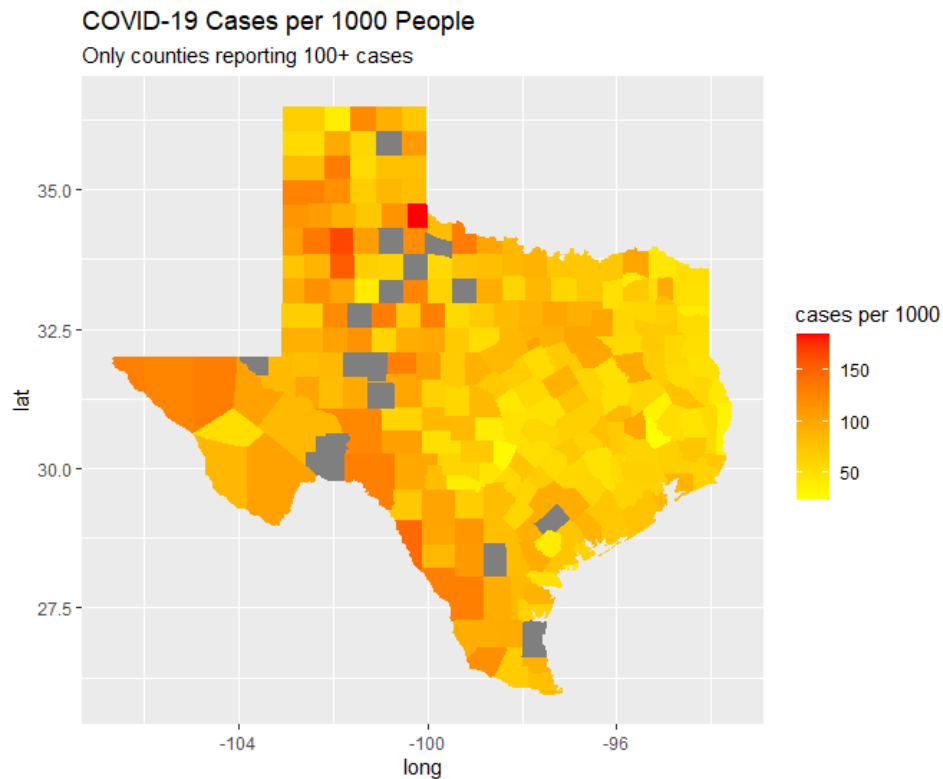


Figure 6: County data illustrating the number of cases found per 1000 people.

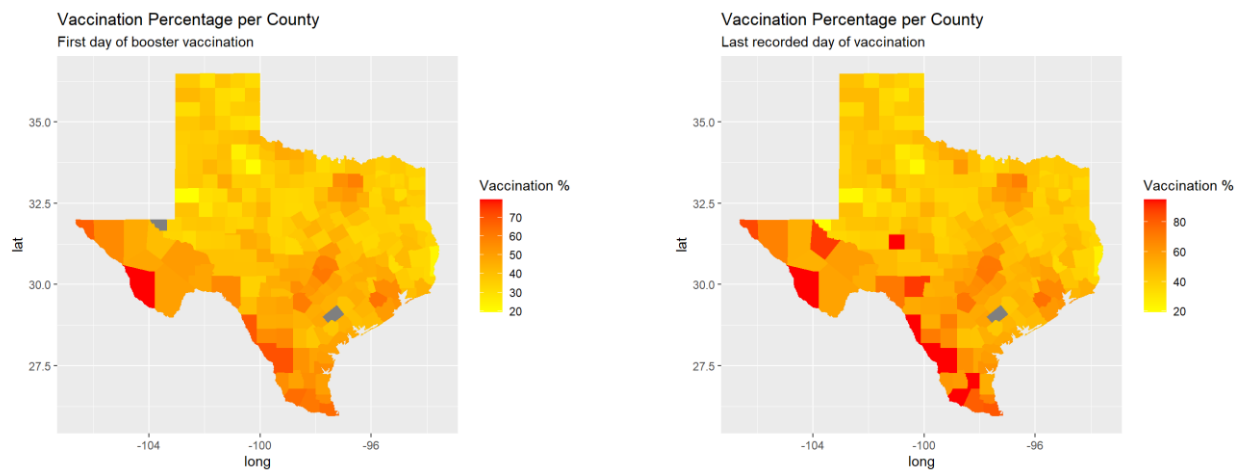


Figure 7: County data illustrating the percent of people that are vaccinated. Figure 8: The percent of people that are vaccinated at the end of COVID.

Here is the data illustrating the percentage of citizens in each county that were vaccinated by the first time the boosters were rolled out to the end of the pandemic. We can see that there is a large correlation between the counties that were more unvaccinated by how lethal the virus was. On figure 7, the darker red indicates more deaths per 1000 cases, and on figures 8 and 9, the darker red indicates the vaccination percentage. One area of note in these three figures is the pan handle which seems to have the highest ratio of cases to deaths, it also so happens to have the lowest vaccination percentage. This could be due to a few reasons: vaccines were not distributed well out west while urban cities and southern counties had easy access. There may not be as many hospitals to treat the virus out west, leading to the small number of citizens that do get sick to have a much higher mortality rate. Another

idea is that the citizens were hesitant to get vaccinated and did not have the immune system to combat the virus, leading to the higher mortality rate. Whatever secondary factors may come into the equation, by observing prior data, plotted onto figure 4, we can see that once the vaccine was at its most effective, the deaths declined sharply, albeit for a short time.

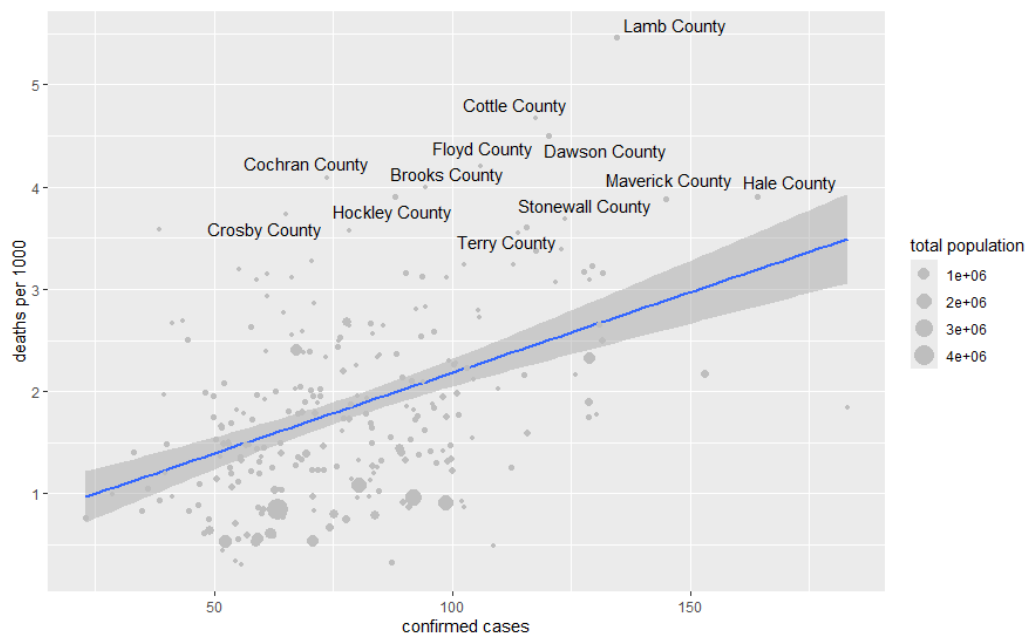


Figure 9: Graph depicting the density of deaths by plotting points with density.

Figure 9 is a graph that illustrates confirmed cases of COVID-19 and deaths per 1,000 individuals in various counties in Texas. The positive relationship between deaths and confirmed cases per 1,000 residents can be seen using the blue trend line from the scatter plot. It is mentioned, however, that rural counties such as Lamb County, Cottle County, and Dawson County have a higher death rate per 1,000 confirmed cases than do more urban, larger counties.

One of the reasons for this disparity is availability of healthcare facilities. Urban counties, despite having more cases overall, are more likely to have hospitals, ICU facilities, and doctors available, which reduces the likelihood of severe cases. Also, as can be seen from Figures 7 and 8, urban areas also enjoy higher vaccination rates, which significantly lower deaths and severity of case. This means that survival is heavily based on healthcare infrastructure and vaccine coverage rather than the total number of cases.

Also, the sizes of the points on the scatter plot represent the total populations of the counties and observe how larger and more populous counties bunch towards the bottom of the death rate scale, whereas smaller counties experience higher death rates. This too reinforces the finding that rural regions are more likely to have worse COVID-19 outcomes since they contain fewer hospitals, fewer immunized individuals, and potentially higher comorbidities.

This graph underscores the importance of targeted public health interventions in rural areas, such as expanding healthcare infrastructure, increasing vaccination outreach, and ensuring timely medical support, to reduce COVID-19 mortality disparities between rural and urban populations.

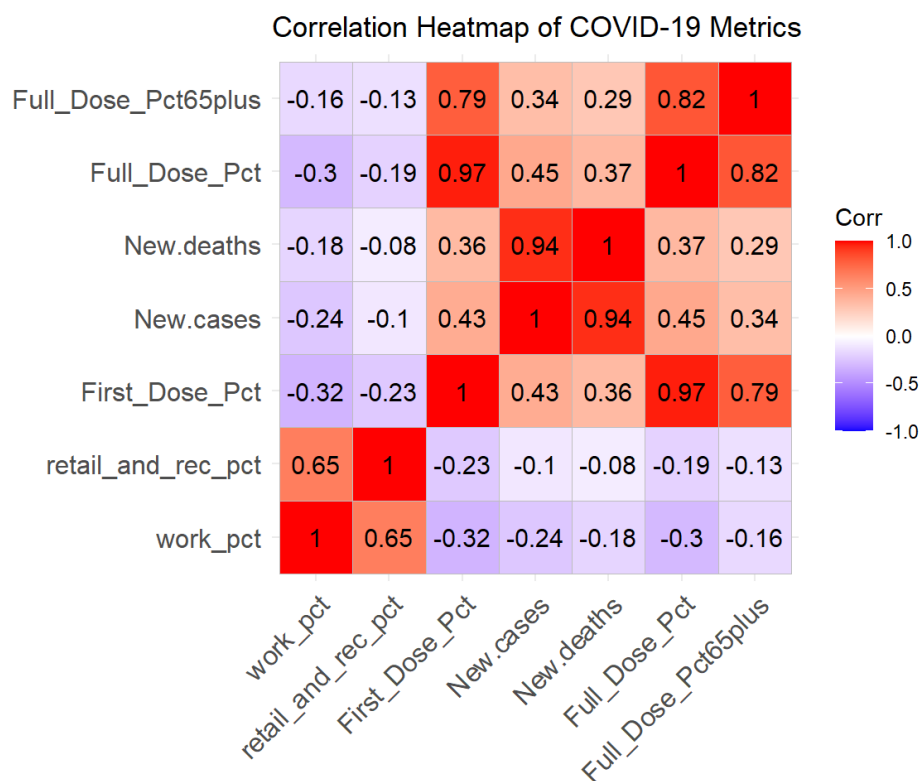


Figure 6: The correlation of vaccines compared to cases and deaths.

The heatmap of figure 8 picks up a couple of less obvious but more reflective correlations that are crucial to better than superficially comprehend COVID-19 dynamics. For instance, the -0.19 negative correlation between Retail_and_rec_pct and First_Dose_Pct suggests that locations with greater retail and recreational activities have reduced first-time closures, which could reflect an exchange-off between public health measures and economic activity. Another weak but meaningful correlation is that of New.cases and Full_Dose_Pct (0.43), indicating that areas with full closure percentages will likely continue to experience increasing new cases because of lagged effectiveness of closures or underlying determinants like population density or healthcare capacity. Additionally, the weak positive correlation between First_Dose_Pct and New.deaths (0.36) implies that initial shutdowns, as measures to reduce transmission, might be implemented as responses to already rising death counts, which reflect the virulence of the outbreak at the time intervention was undertaken. These more subtle correlations work to reflect the intricacy of the impact of the pandemic, toward the imperative of varied response based on considerations not merely of short-term health measures but of longer-term economic and social impacts.

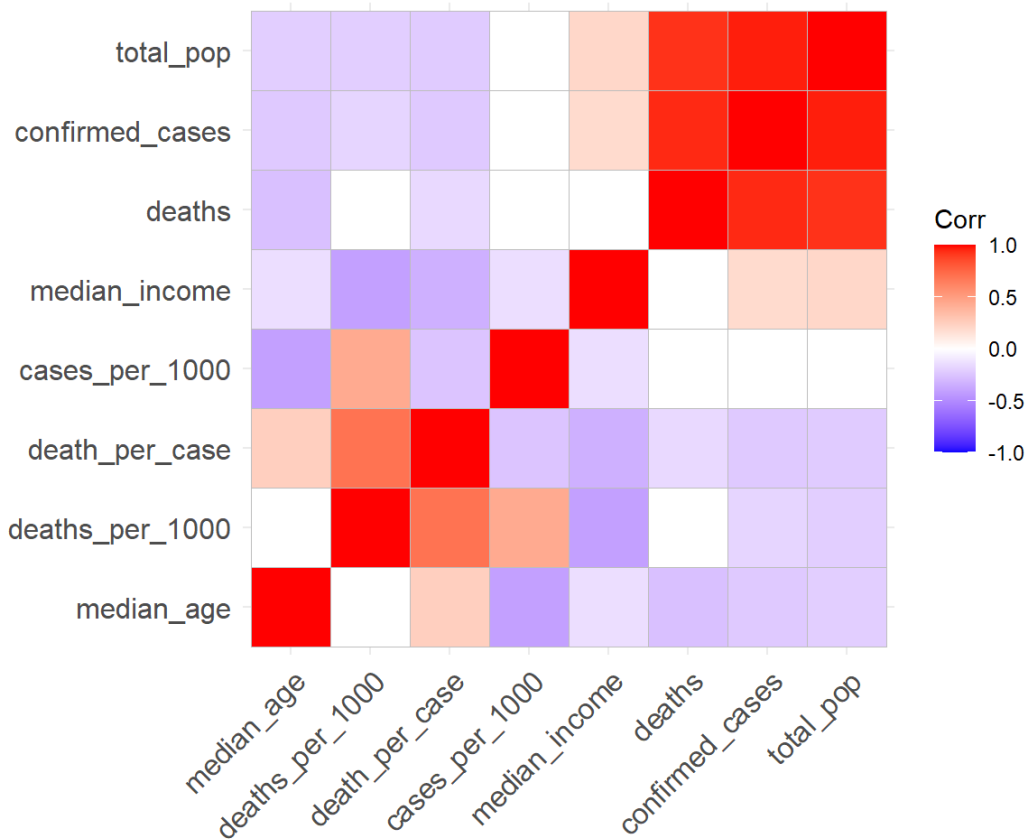


Figure 7: The correlation of population, income, and age to the chance getting sick or death.

The correlation heatmap provides us with a sense of the inter-relationships among various COVID-19 measures and demographic factors. There are strong positive correlations (red) among the total population, confirmed cases, and deaths since one would anticipate this given that denser populations have more deaths and cases. Death per case and deaths per 1,000 also correlate highly positively, indicating that regions with higher numbers of deaths per case would also tend to have a higher rate of total death proportionally to the population. Median age also correlates highly positively with deaths per 1,000, indicating that higher median-aged populations may have experienced higher mortality rates because of COVID-19. Meanwhile, median income is weakly negatively correlated with cases per 1,000, which could be interpreted as richer areas having lower case rates due to possibly more access to medical care or being able to work from home. Several weaker or negative correlations (blue) show that variables like median income and population are perhaps not direct indicators of COVID-19 severity if one considers fatality rates. In general, these interactions highlight the role of demographic factors in explaining the impact of the pandemic at the regional level.

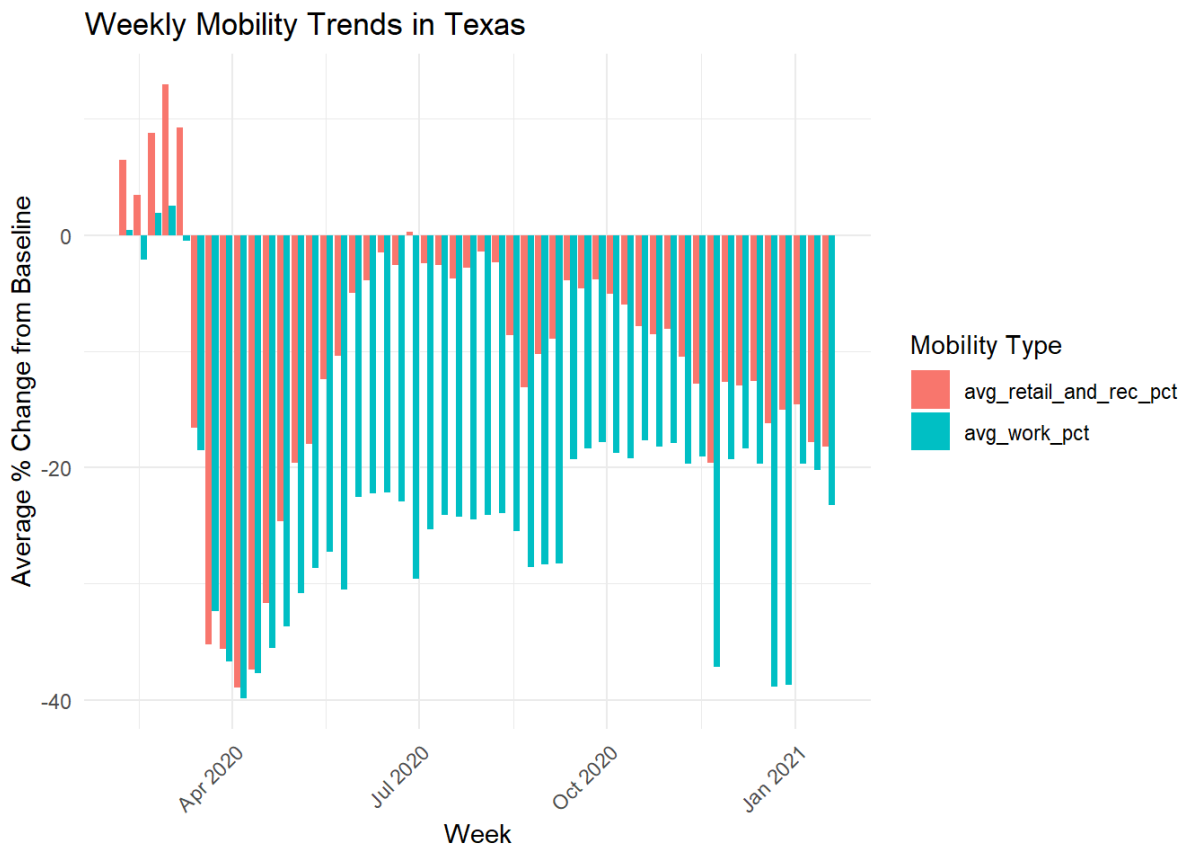


Figure 10: Weekly mobility trend, highlighting trends in working from home and retail and recreation.

The graph plots weekly mobility trends in Texas by presenting the percent change from baseline of two types: retail and recreation (red bars) and workplaces (blue bars). The trend is apparent, with the intense decline in mobility in March-April 2020, the time when the initial wave of COVID-19 lockdowns occurred. Work-related mobility declined more than retail and recreation, indicating that while most workplaces adopted remote work, some retail stores were open subject to restrictions. Later in 2020, both forms of mobility experienced gradual restoration but never to pre-pandemic levels. Retail mobility experienced more fluctuation, likely echoing evolving public health policy and reopening phases, but workplace mobility was relatively stable, indicating the persistence of remote working policies. Another key observation is the steep drop in workplace mobility during late 2020 and early 2021, which aligns with holiday periods and reimposed restrictions as COVID-19 cases increased.

The histogram was chosen as the most appropriate visualization of the data because it is effective in illustrating trends over time while making a comparison between the two mobility types evident. The stacked bars render the contrast between work and retail mobility per week readily readable in terms of behavioral pattern differences. The bar chart is also better suited to time-segmented data, which avoids artificial continuity. It also effectively plots the negative data points in a clear and readable way.

5 Recommendations

Based on data analysis, Texas policymakers must take a proactive and data-driven approach when addressing future public health crises. The insights gained from examining vaccination rates, hospitalization trends, social distancing measures, and demographic factors reveal key areas for intervention. By analyzing these charts and variables, Texas can ensure better preparation and response for future outbreaks. These recommendations are designed to help state-level policymakers in Texas develop more effective public health strategies, ensuring better preparedness and response for future outbreaks.

➤ 1. Enhance Targeted Vaccination Efforts

One of the strongest findings in the analysis is the clear correlation between low vaccination rates and high mortality rates across Texas counties. Rural and less densely populated areas showed significantly lower vaccination rates, yet they had a higher proportion of deaths per confirmed COVID-19 case. The data suggests that these communities faced barriers to vaccination, whether due to accessibility, vaccine hesitancy, or misinformation. Launching targeted public awareness campaigns addressing vaccine safety, efficacy, and the importance of booster shots will help encourage participation, particularly among hesitant populations.

➤ 2. Monitor and Respond to Variant Surges More Rapidly

Each new COVID-19 variant resulted in a rapid spike in cases before interventions were able to keep up, demonstrating the importance of early detection and response. A lag in policy adjustments—whether in vaccine distribution, mask mandates, or mobility restrictions—allowed new variants to spread unchecked before effective containment measures were introduced, and this was the major issue. To prevent this in future outbreaks, the Texas government should establish a faster system that can track mutations in real-time. By integrating real-time epidemiological data with policy decision-making, interventions can be implemented earlier, preventing unnecessary hospitalizations and deaths. Policymakers should also consider preemptively increasing public health messaging and hospital readiness when new variants are detected, rather than waiting for case surges to dictate action.

➤ 3. Maintain Social Distancing Policies During High Transmission Periods

The analysis of mobility data showed that increases in movement, particularly in retail and workplaces, were closely followed by COVID-19 surges. While economic activity is crucial, failing to regulate public interactions during periods of high transmission leads to unnecessary outbreaks. The data suggests that temporary and targeted social distancing measures, such as mask mandates, capacity limits, and work-from-home policies, could significantly reduce transmission rates. Rather than implementing blanket lockdowns, policymakers should use the data-driven approach to identify high-risk areas and **temporarily reinstate restrictions only where necessary**. This approach balances economic stability with public health, ensuring businesses can continue operating while minimizing health risks.

➤ 4. Increase Public Awareness on Vaccine Boosters

Vaccination alone was not enough to defend against all deaths, however with boosters, intensive care and mortality rates declined sharply. The analysis highlights the direct link between booster rollout and the decline in deaths, showing the immunity reinforcement is the best course of action to maintain public health. However, many communities exhibited booster hesitancy, often due to misconceptions or lack of accessibility. Texas could launch a campaign of data driven explanations towards those who are at higher risk yet still hesitant.

➤ 5. Address Healthcare Disparities in Rural Areas

Rural counties had the highest mortality rate and the lowest vaccinations rates, which could be a result of systematic healthcare disparities, such as limited access to well-equipped hospitals or delayed medical attention. Policymakers should increase funding for rural healthcare by incentivizing medical professionals to work in these more rural areas through salaries, benefits, and working conditions. Expanding on mobile clinics, emergency response units in rural areas would ensure faster and more effective treatment, preventing unnecessary deaths.

6 Conclusion

This study fully addresses the first question by presenting evidence-based findings with respect to the effectiveness of social distancing and vaccination policies in curbing COVID-19 infection rates, death, and hospitalization rates. The findings reveal that while social distancing initially led to a decrease in infection rates, its effect was short-lived, with infections beginning to surge once more after restrictions were relaxed. On the other hand, vaccination saw a more lasting effect in reducing infection rates, with a notable decline in the occurrence of serious illness and death. As such, for example, after the launch of the vaccine roll-out in early 2021, there was a dramatic drop in reported infections in Texas, despite the easing of movement restrictions. This observation would suggest that vaccination presented a superior long-term defense against the virus in comparison with social distancing. In addition, while the Omicron variant did lead to an early 2022 surge in infections, death and hospitalization rates were less than during previous spikes, thus further supporting that vaccination was the key in precluding serious illness during subsequent outbreaks, in comparison with social distancing policies.

The empirical data on the impact of social distancing and vaccination on hospitalization and mortality rates is consistent in showing that vaccination is the more effective measure. Before vaccines were widely available, hospitalization rates tracked closely with rises in infection rates, thus putting significant pressure on healthcare systems. After more widespread vaccine distribution, however, there was a significant reduction in ICU admissions. For instance, during the Delta variant wave in early 2021, Texas saw over 3,600 new ICU admissions per week at its peak. In sharp contrast, by late 2021, after booster shots were introduced, ICU admissions during the Omicron surge ran at a markedly lower rate despite a large number of cases. This information serves to underscore that case

severity was reduced through vaccination, in turn reducing undue pressure on hospitals. Social distancing, though yielding temporary hospital relief, was not a viable long-term measure in that hospitalization rates rose whenever restrictions were eased.

The evidence suggests that vaccination has greatly improved survival in cases with the infection. Areas with high rates of vaccination consistently reported fewer death rates per 1,000 infections, thus illustrating that vaccines contributed significantly towards lessening mortality. In Texas, for instance, the death rate per case fell from more than 1.5% in 2020 down to around 0.5% by the latter stages of 2021, after at least one vaccine dose was administered to most members of the population. In addition, intensive care unit admissions were decreased among vaccinated persons, further supporting that vaccines help prevent serious disease, even during breakthrough infections. These findings greatly inform policymakers since they reveal that immunization programs represent the most effective means of lessening death from COVID-19.

The studies suggest that the long-term effect of vaccination is stronger than that of social distancing in reducing the impact of the pandemic. While social distancing was helpful in early stages, it was not a long-term solution, with diminishing efficacy after restrictions were removed. In contrast, vaccination provided long-term protection, greatly relieved intensive care unit loads, and significantly reduced mortality. The findings suggest that policymakers should focus on expediting vaccine distribution, starting booster campaigns, and prioritizing variant studies over extended social distancing restrictions. However, short-term distancing methods might still be necessary in reaction to the introduction of new, highly contagious mutants, particularly when there is a risk of overloading hospitals. The introduction of new mutants is a serious threat, especially with the limitation of available information; however, waiting for effective treatments is a reasonable approach. Utilizing real-time information in guiding public health responses, Texas and other states can optimize the efficacy, specificity, and sustainability of subsequent responses to the pandemic.

7 List of References

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8 Appendix

8.1 Student Contributions

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